EVIDENCE GUIDE

Box3D Metal Compatibility and Validation

The exact GPU-resident surface, explicit CPU boundary, differential evidence, and completed acceptance matrix.

APPLE SILICON / METAL COMPUTE / BOX3D OVERLAY

Compatibility contract

The reference is unmodified Box3D commit `47d7f7cc7e091142c08d11dc7d2e493c5d34f536`. Metal is opt-in and tolerance-equivalent, not bit-identical across platforms.

GPU-resident surface	Modes
Fused integration	Velocity and position across all substeps
Convex contacts	Normal, friction, tangent velocity, twist, rolling, restitution
Mesh contacts	Scalar multi-manifold solve and restitution
Distance joints	Rigid, spring, limit, motor, static bodies
Parallel joints	Soft alignment, torque limiting, static bodies
Overflow	Serial deterministic contact and mixed supported-joint order
Pair traversal	Experimental exact-order raw dynamic-tree candidates

Explicit CPU boundary

- Broad-phase tree mutation, pair filtering, narrow phase, and manifolds
- Contact and joint preparation
- Events, islands, sleeping, and CCD
- Recording, queries, topology mutation, and public API calls
- Filter, motor, prismatic, revolute, spherical, weld, and wheel solving
- Any joint requesting reaction-threshold events

Unsupported constraints keep the reference CPU solve for that step. The position-only Metal stage may still run if its threshold is met.

Differential evidence

Matching CPU and Metal worlds run for multiple steps/substeps. Transforms and linear/angular velocities are compared after each scenario. Values below are observed maxima, not universal bounds.

Scenario	Transform	Velocity
Convex friction	4.77e-7 transform	3.98e-6 velocity
Mesh contacts	4.77e-7	3.60e-6
Distance modes/overflow	1.07e-6	8.27e-5
Mixed distance/parallel	7.45e-9	1.79e-7
Mixed joint overflow	1.82e-6	3.29e-4
Static-body joints	4.47e-8	-

Acceptance matrix

Debug and Release full suites; CPU-only Release; double-precision Metal differential tests; AddressSanitizer and UndefinedBehaviorSanitizer; warning-as-error build; shared dylib demo; CMake install audit; and clean-clone patch reconstruction.

Failure semantics. Initialization failure returns false and leaves the world usable. Unsupported work increments fallback counters. Disabling Metal releases resources without destroying the world.